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MEDICAL DEVICES FOR AGENT DELIVERY AND RELATED METHODS OF USE

Abstract

A device may be configured to deliver an agent to a target tissue via pressurized fluid and the device may include a catheter including a lumen; an enclosure configured to store an agent, receive a pressurized fluid, and release a combination of the pressurized fluid and the agent; and a valve downstream of the enclosure to receive the combination of the pressurized fluid and the agent from the enclosure. The valve may have a first configuration preventing flow of the combination through the lumen and a second configuration permitting flow of the combination through the lumen.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application is a continuation of U.S. patent application Ser. No. 17/109,434, filed on Dec. 2, 2020, which claims the benefit under 35 U.S.C. § 119 to U.S. Provisional Patent Application No. 62/943,060, filed on Dec. 3, 2019, each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to medical systems and devices for delivering pressurized fluids/agents, and more particularly, to methods and tools for controlling delivery of the fluid/agent at an appropriate pressure and flow rate.

BACKGROUND

[0003] In certain medical procedures, it may be necessary to stop or minimize bleeding internal to the body. For example, an endoscopic medical procedure may require hemostasis of bleeding tissue within the gastrointestinal tract, for example in the esophagus, stomach, or intestines.

[0004] During an endoscopic procedure, a user inserts a sheath of an endoscope into a body lumen of a patient. The user utilizes a handle of the endoscope to control the endoscope during the procedure. Tools are passed through a working channel of the endoscope via, for example, a port in the handle, to deliver treatment at the procedure site near a distal end of the endoscope. The procedure site is remote from the operator.

[0005] To achieve hemostasis at the remote site, a hemostatic agent may be delivered by a device inserted into the working channel of the endoscope. Agent delivery may be achieved through mechanical systems, for example. Such systems, however, may require numerous steps or actuations to achieve delivery, may not achieve a desired rate of agent delivery or a desired dosage of agent, may result in the agent clogging portions of the delivery device, may result in inconsistent dosing of agent, or may not result in the agent reaching the treatment site deep within the GI tract. The current disclosure may solve one or more of these issues or other issues in the art.

SUMMARY OF THE DISCLOSURE

[0006] Examples of the present disclosure relate to, among other things, agent delivery devices. Each of the examples disclosed herein may include one or more of the features described in connection with the disclosed examples.

[0007] A device may be configured to deliver an agent to a target tissue via pressurized fluid and the device may include a catheter including a lumen; an enclosure configured to store an agent, receive a pressurized fluid, and release a combination of the pressurized fluid and the agent; and a valve downstream of the enclosure to receive the combination of the pressurized fluid and the agent from the enclosure. The valve may have a first configuration preventing flow of the combination through the lumen and a second configuration permitting flow of the combination through the lumen.

[0008] Any of the systems and devices disclosed herein may have any of the following features. A first fluid input may be configured to release pressurized fluid from a pressurized fluid container into the lumen upon coupling the container to the first fluid input. The valve may be within a housing, and the housing may include a handle. The agent may be a powdered medicament. The valve may include a chamber, and the chamber may be fluidically connected to the lumen via an

output channel. The valve may also include a rod including a plunger at a distal end of the rod, and the rod may extend through the chamber and may be configured to prevent fluid flow to the output channel in a first configuration and to allow fluid flow to the output channel in a second configuration. The valve may further include a slider slidably coupled to the chamber and including a slot, and the rod may extend through the slot, and the slider may be configured to transition the valve from a first configuration to a second configuration by movement of the slider. The valve may also include a biasing member coupled to the rod, and the biasing member may be configured to bias the rod towards the first configuration of the valve. The slider may include a slide on which the rod rests, the slide having a first portion including a curved surface and a second portion including a planar surface. The rod may include an end extending radially outward relative to an adjacent portion of the rod, and wherein the end of the rod prevents the rod from moving through the slot when the end of the rod contacts the slider. The end of the rod may slidably engage the slider.

[0009] The valve may include a chamber, and the chamber may be fluidically connected to the lumen via an output channel. The valve may also include a rod including a plunger at a first end of the rod and a cam-engaging surface at a second end of the rod, wherein the rod extends through the chamber and is configured to prevent fluid flow to the output channel in the first configuration and to allow fluid flow to output channel in the second configuration. The valve may further include a cam engaging the cam-engaging surface; a pinion fixedly coupled to the cam and comprising a first plurality of gears; a rack comprising a second plurality of gears, wherein the second plurality of gears are configured to mate with the first plurality of gears; and a biasing member coupled to the rod, wherein the biasing member is configured to bias the rod towards the first configuration of the valve. The cam may be pear shaped, the rod may be U-shaped, and the cam-engaging surface may include a wheel. A trigger may be coupled to the rack, wherein actuation of the trigger moves the rod to transition the valve from the first configuration to the second configuration. Movement of the rack in a first direction may be configured to move the pinion and the cam to transition the cam from a first position in which a central longitudinal axis of the cam is substantially parallel to a central longitudinal axis of the rack to a second position in which the central longitudinal axis of the cam is transverse to the central longitudinal axis of the rack; and movement of the rack in a second direction may be configured to move the pinion and the cam from the second position to the first position. The valve may include a chamber, wherein the chamber includes a distal opening fluidically connecting an interior portion of the chamber to the output channel; and a butterfly valve positioned within the distal opening. The butterfly valve may be configured to prevent fluid flow to the output channel in the first configuration of the valve and to allow fluid flow to the output channel in the second configuration of the valve, and rotation of the butterfly valve may move the butterfly valve to transition the valve from the first configuration to the second configuration.

[0010] The lumen may be a first lumen and the valve may include a chamber, wherein the chamber includes a distal opening fluidically connecting an interior portion of the chamber to the output channel; and a ball valve positioned within the distal opening and including a second lumen extend through the ball valve. The ball valve may be configured to prevent fluid flow to the output channel in the first configuration of the valve and to allow fluid flow through the second lumen to the output channel in the second configuration of the valve, and rotation of ball valve may transition the ball valve from the first configuration to the second configuration. A housing may include a handle, wherein the handle includes an interior portion configured to receive a fluid container.

[0011] A delivery device may be configured to deliver an agent to a target tissue via pressurized fluid. The delivery device may include: a catheter including a lumen, a proximal end, and a distal end; an enclosure configured to store an agent, receive a pressurized fluid, and release a combination of the pressurized fluid and the agent; and a first fluid input upstream of the enclosure. The first fluid input may be configured to release pressurized fluid from a pressurized fluid container toward

the enclosure upon coupling the container to the first fluid input. In some examples, the pressurized fluid and agent may flow from the enclosure to the distal end of catheter upon coupling the container to the first fluid input. A total volume of pressurized fluid within the container may be configured to deploy a first amount of the agent through the distal end of the catheter.

[0012] In other examples, a method for controlling a fluid delivery to a body of a patient is disclosed. The method may include fluidically connecting an enclosure to a catheter, wherein the enclosure includes an agent within an interior portion of the enclosure and is configured to feed the agent into a lumen of the catheter; moving a distal end of the catheter to a target tissue site, the catheter including a lumen extending longitudinally through the catheter, and wherein the lumen is configured to receive a pressurized fluid from a container through a first input at a position proximally of the enclosure; coupling the container to the first input, wherein the container releases pressurized fluid stored within the container into the first input upon coupling the container to the first input; and contacting an actuator of a valve to open a fluid pathway from the first input to a distal opening of the lumen, causing a fluid and the agent to be released out of the distal end of the catheter. In some examples, the method may further include compressing a spring within the valve assembly; and breaking a fluid seal between an input channel and an output channel of the valve, wherein breaking the fluid seal fluidically connects the first input with the distal opening of the lumen.

[0013] It may be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed. As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Throughout the drawings, The term “exemplary” is used in the sense of “example,” rather than “ideal.” The term “distal” refers to a direction away from an operator, and the term “proximal” refers to a direction toward an operator. The term “approximately,” or like terms (e.g., “substantially”), includes values $\pm 10\%$ of a stated value. The term “distal” refers to a portion farthest away from a user when introducing a device into a patient. By contrast, the term “proximal” refers to a portion closest to the user when placing the device into the patient. Proximal and distal directions are labeled with arrows marked “P” and “D”, respectively, throughout the figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.

[0015] FIG. 1 is a side view of a delivery system according to an exemplary embodiment.

[0016] FIG. 2 is a side view of a delivery system according to an exemplary embodiment.

[0017] FIGS. 3A and 3B are cross-sectional views of a valve assembly in first and second configurations, respectively, according to an exemplary embodiment.

[0018] FIGS. 4A and 4B are top views of a valve assembly in first and second configurations, respectively, according to an exemplary embodiment.

[0019] FIGS. 5A and 5B are cross-sectional views of a valve assembly in first and second configurations, respectively, according to an exemplary embodiment.

[0020] FIGS. 6A and 6B are cross-sectional views of a valve assembly in first and second configurations, respectively, according to an exemplary embodiment.

[0021] FIGS. 7A and 7B are cross-sectional views of a valve assembly in first and second

configurations, respectively, according to an exemplary embodiment.

[0022] FIG. **8** is a cross-sectional side view of a handle assembly including a valve assembly, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0023] An agent delivery device may be configured to house an agent at a distal portion of the delivery device and may be configured to control a rate at which an agent and a fluid leave the delivery device at a single location. The delivery device may be configured to house single component agents or multi-component agents. In the case of multi-component agents, the delivery device may be configured to allow for mixing of the components, prior to delivery from the device. Various mechanisms may be utilized in order to pre-pressurize a chamber of the agent and actuate the delivery device to deliver the agent while limiting the number of valves required in the device. These mechanisms may include pneumatics, wires, tubes, valves, or any suitable combination thereof.

[0024] Referring to FIG. **1**, a delivery system **100** according to an embodiment is shown. Delivery system **100** may include a catheter **102**, a enclosure **104** containing an agent **112**, an enclosure adapter **105**, a fluid container **106**, a regulator **107** including a fluid source input **109**, and a filter **110**. In some examples, fluid container **106** may release pressurized fluid into regulator **107** when coupled to regulator **107**. For example, input port **109** may be configured to pierce fluid container **106** when fluid container **106** is coupled to regulator **107**, releasing pressurized fluid into regulator **107**.

[0025] With reference to FIG. **1**, fluid container **106** is configured to contain a fluid, such as a gas, e.g., carbon dioxide or any other gas or other fluid known in the art. While shown as a cylinder, fluid container **106** may be any shape, such as a torpedo-shape, a sphere, or any other shape known in the art and used for storing fluid. For example, fluid container **106** could be a carbon dioxide tank or cylinder typically found in medical settings, such as a hospital, or a smaller, portable cartridge. In some examples, fluid container **106** may be fluidically connected to a fluid source separate from delivery system **100**, such as via a fluid catheter that may fluidically connect to a stationary fluid source in an operating room. Fluid container **106** may include one or more outer walls defining one or more inner chambers (not shown), the inner chamber(s) configured to contain the fluid. The walls of fluid container **106** may be formed of any material suitable for containing the fluid, such as but not limited to a metal alloy, a ceramic, or other material known in the art. The fluid contained in the inner chamber of fluid container **106** may be under pressure. Accordingly, the walls are formed of a material and/or a thickness suitable to contain the fluid at a pressure of, for example, at least approximately 1000 pounds per square inch (PSI), or approximately 850 PSI. For example, gases which may be contained in fluid container **106** may include carbon dioxide (CO₂) having a vapor pressure of approximately 2,000-8,000 kPa at typical device temperatures, or nitrogen (N₂) having a vapor pressure less than 40 MPa at typical device temperatures. It will be understood that these gases are examples and are not limiting to the types of gases contained in fluid container **106**.

[0026] With continued reference to FIG. **1**, fluid container **106** is attached to regulator **107** at regulator input opening **109**. Regulator **107** includes a body **108** (including an input opening **109** and an output opening **111**, each for communication to external environment). Input opening **109** of regulator **107** may be configured to pierce fluid container **106**. Output opening **111** may be configured to fluidically couple to a proximal end of catheter **102**. Body **108** of regulator **107** may be any material known in the art, including but not limited to a metal alloy, a ceramic, and/or a resin. Regulator **107** may be configured to change to flow rate of fluid exiting fluid container **106** such that fluid enters a distal portion of catheter **102** at a selected flow rate. In some examples, fluid may enter a distal portion of catheter **102** at a flow rate in the inclusive range of approximately four standard liters per minute to twelve standard liters per minute of fluid flow. In some examples, delivery system **100** may be configured to operate with fluid flowing through a distal portion of

catheter **102** at a flow rate in the inclusive range of between approximately five standard liters per minute to ten standard liters per minute. Fluid container **106** may be coupled to regulator **107** via a pull-cord coupling mechanism or a pump mechanism. In some examples, a pull-cord coupling mechanism may be included in system **100** and may include an actuator including a cord and a pin, and the pin may engage a carbon dioxide cartridge (or other fluid cartridge) to release fluid from the cartridge when a user pulls the cord. For example, the pin may puncture the cartridge and result in pressurized fluid flowing in system **100**. In other examples, a pump mechanism may be included in system **100** and may be configured to supply pressurized fluid to system **100**, for example by supplying pressurized gas in system **100**. In some examples, system **100** may not include regulator **107** and container **106** may be coupled directed to catheter **102** via an input opening similar to input opening **109**.

[0027] A distal portion of catheter **102** (a portion distal to regulator **107**) may supply fluid under pressure from fluid container **106** and regulator **107** to enclosure **104** and enclosure adapter **105**. Enclosure **104** may be cylindrical and may include an interior cavity **118** and a funnel portion **119** at an end of enclosure **104**. Enclosure **104** may be configured to store an agent **112**, such as a powder or liquid medicament, within interior cavity **118** and interior cavity **118**, may be fluidically connected to adapter **105**. Funnel portion **119** may include tapered surfaces forming a portion of interior cavity **118** that lead to an opening **123** at an end of enclosure **104**. Funnel portion **119** may be configured to direct agent **112** through opening **123** into adapter **105**. In some examples, enclosure **104** may be configured to gravity feed agent **112** into adapter **105**. Enclosure **103** may be any suitable material known in the art. In some examples, enclosure **103** may be made of a transparent material.

[0028] Opening portion **123** may be configured to couple to adapter **105**. Adapter **105** may include an interior cavity (not shown) fluidically connecting three separate openings. Two of the three separate openings may be configured to couple to portions of catheter **102**, and one of the three separate openings may be configured to couple to opening **123** of enclosure **104**. An opening of adapter **105** may be configured to removably couple to enclosure **104**. Adapter **105** may be made of metal, polymer, or any other suitable material known in the art.

[0029] Catheter **102** may be cylindrical and may include a lumen extending along its central longitudinal axis. In some examples, catheter **102** may include a proximal portion **122** and a distal portion **121**. Proximal portion **122** of catheter **102** may be coupled to adapter **105** and may fluidically connect regulator **107** with adapter **105**. Distal portion **121** of catheter **102** may be coupled to adapter **105** at a proximal end of distal portion **121**, and distal portion **121** may extend to a distal opening **115**. Catheter **102** may be made of any material, for example reinforced rubber or a suitable plastic, that allows catheter **102** to withstand the pressures of the fluid, while simultaneously allowing for unrestricted movement of catheter **102**. Distal portion **121** may be flexible and be configured to bend to facilitate movement through a body lumen of a patient, e.g. a gastrointestinal tract. Proximal portion **122** of catheter **102** may include a filter **110** positioned within the lumen of catheter **102**. Filter **110** may be configured to allow fluid flow from the regulator **107** distally through the lumen of catheter **102**, and may also be configured to prevent movement of agent **112** proximally through catheter **102**. In some examples, filter **110** may be configured to restrict fluid flow through catheter **102** by narrowing the diameter of the interior lumen of catheter **102**. System **100** may provide a means to deliver agent **112** to a target tissue without the need of any valves within catheter **102**, which may reduce procedure time and may simplify a procedure of delivery of an agent **112** to tissue.

[0030] In operation, a user may first insert an agent **112** into enclosure **104** of delivery system **100**. The user may then position distal opening **115** of catheter **102** proximate to target tissue of a patient, for example target tissue within a body lumen of a patient. Once distal opening **115** is positioned proximate to or at the target tissue, the user may couple fluid container **106** to regulator **107**. By coupling fluid container **106** to regulator **107**, input port **109** may pierce or otherwise

initiate fluid flow from fluid container **106** through regulator **107** and into catheter **102**. Fluid may then flow from regulator **107** through proximal portion **122** of catheter **102** and into adapter **105** and enclosure **104**. When fluid flows into adapter **105** and enclosure **104**, agent **112** may be moved in the direction of fluid flow and carried through catheter **102** to distal opening **115**. Agent **112** may then be deployed through distal opening **115** and propelled towards the target tissue via the fluid flow. Since the fluid flow is initiated when fluid container **106** is coupled to regulator **108**, a user does not need to actuate a valve or otherwise adjust the deployment of agent via fluid flow from fluid container **106**. The actuation of system **100** through coupling fluid container **106** to regulator **107** (for example, piercing a carbon dioxide container by coupling it to regulator **107**) provides a means to deliver agent **112** to target tissue without the need for valving and produces a single shot of agent **112** to target tissue. For example, system **100** may be configured to receive a single shot of pressurized fluid supplied by fluid container **106**, such as a specific volume of carbon dioxide released from fluid container **106** in order to deliver a specific amount of agent **112** to distal opening **115**. By providing the user with a device that may deliver a measured amount of agent by connecting a fluid container **106** with a specific amount of fluid, additional steps of actuating a valve assembly or selectively releasing liquid into a device via an actuator may be eliminated, and may save the user time during an operation. In some examples, system **100** may not include regulator **107**. In some examples, all of or the majority of agent **112** may be moved out of enclosure **104** when fluid container **106** is coupled to regulator **108**, e.g., each enclosure **104** may be a single dosage of a therapeutic agent. The volume of fluid enclosure **106** may be configured to deploy a selected amount of agent **112** using system **100**.

[0031] FIG. 2 shows an alternative embodiment of a delivery system **200** substantially similar to delivery system **100**. Delivery system **200** may include catheter **202** including distal portion **222**, proximal portion **223**, and distal opening **215**; adapter **205**; fluid container **206**; regulator **207** including regulator body **208**, regulator input opening **209** and output opening **211**; enclosure **204**; agent **212**; and filter **210**. Any of the features described herein above regarding delivery system **100** may be included in delivery system **200**. Delivery system **200** may also include a valve **220** at a distal portion of catheter **202**, e.g. any portion of catheter **202** distal of adapter **205**. Valve **220** may be fluidically connected to an interior lumen of catheter **202**. In some examples, valve **220** may be coupled to distal portion **222** of catheter **202** and a distalmost portion **221** of catheter **202**. In other examples, valve **220** may be just distal of adapter **205**.

[0032] In operation, a user may first couple fluid container **206** to regulator **207** to pressurize delivery system **200** with pressurized fluid, such as pressurized carbon dioxide. Delivery system **200** may be configured to withstand cylinder pressure with valve **220** keeping the fluid pressure and agent **212** contained within delivery system **200**. Once delivery system **200** has been pressurized with fluid by coupling fluid container **206** to regulator **207**, the user may then selectively release the pressurized fluid and agent **212** from delivery system **200** by actuating valve **220**. In some examples, actuation of valve **220** may result in release of fluid and/or agent **212** through distalmost portion **221** of catheter **202** and out of distal opening **215**. By providing valve **220** at a distal portion of catheter **202**, a user may selectively delivery agent **212** to a target tissue and may allow the user to deliver agent **212** multiple times to the same target area or different target areas. Valve **220** may have any number of mechanisms for regulating the release of fluid from delivery system **200** which are discussed herein below.

[0033] FIGS. 3A and 3B show cross-sectional views of an exemplary valve assembly **300**. Valve assembly **300** may be used as valve **220** in FIG. 2. Valve assembly **300** may be configured to operate under a constant fluid pressure feed into input channel **351**. Valve assembly **300** may include a valve chamber **323**, an input channel **351** including input lumen **350**, an output channel **353** including output lumen **352**, a rod **325** including a plunger **327** at a distal end of rod **325**, a biasing member **331**, and a slide **321**. Valve assembly **300** may be contained within a holding apparatus (not shown) that may be any suitable shape. For example, valve assembly **300** may be

incorporated into a user for a hold during operation (shown in FIG. 8).

[0034] Chamber **323** may be cylindrical and may include a cover portion **333** at a proximal end of chamber **323** and a distal opening **390** fluidically connecting chamber **323** with output channel **353**. Chamber **323** may include a tapered distal portion **358** with a conical interior surface **359** tapering towards a central longitudinal axis of chamber **323** and towards distal opening **390**. Input channel **351** may be cylindrical and may fluidically connect to chamber **323** at an opening **354** at a proximal portion of chamber **323**. In some examples, distal portion **222** of catheter **202** (shown in FIG. 2) may be fluidically coupled to input channel **351**, and distalmost portion **221** of catheter **202** (shown in FIG. 2) may be fluidically coupled to output channel **353**. Input channel **351** may be positioned above output channel **353** to facilitate fluid flow (such as by allowing gravity to exert force onto the fluid to push the fluid towards output channel **353**) from input channel **351** through chamber **323** to output channel **353**.

[0035] Rod **325** may be cylindrical and may extend through chamber **323**. In some examples, rod **325** may extend along a central longitudinal axis of chamber **323**. Plunger **327** may be coupled to a distal end of rod **325**. Plunger **327** may be conical and may extend radially-outward from a central longitudinal axis of rod **325**. In some examples, rod **325** may be tapered such that the distance between a central longitudinal axis of rod **325** and the radially-outer surface of plunger decreases as plunger **327** extends distally. Plunger **327** may be rubber, hard plastic, or any other suitable material known in the art. Plunger **327** may be configured to form a fluid-tight seal with chamber **323** at distal opening **390**. In some examples, rod **325** may be configured to translate distally within chamber **323** until plunger **327** contacts an interior surface of chamber **323** and forms a fluid tight seal preventing fluid from flowing through distal opening **390**. Proximal end **329** of rod **325** may be tapered and may extend radially outward from a central longitudinal axis of rod **325** as proximal end **329** extends proximally. Rod **325** may extend through an opening **371** (shown in FIGS. 4A and 4B) in slider **321** and also through an opening in cover **333**. Proximal end **329** of rod **325** may be tapered so as to prevent rod **325** from moving through opening **371**. Rod **325** may move proximally (up in the Figures) through opening **371**, such as when proximal end **329** is translated along a proximally-facing surface **355**, **357** of slider **321**.

[0036] Biasing member **331** may extend circumferentially around rod **325** and may be coupled to cover **333** via a coupler **337**. Coupler **337** may fixedly couple a proximal end of biasing member **331** to cover **333**. A distal end of biasing member **331** may be fixedly coupled to plunger **327** and may contact a proximal-facing surface **361** of plunger **327**. In some examples, biasing member **331** may be a spring. Rod **325** may extend through a central longitudinal axis of biasing member **331**. Biasing member **331** may exert a force on rod **325** pushing rod **325** towards distal opening **390**. In an extended state (shown in FIG. 3A), biasing member **331** may exert a force on rod **325** such that plunger **327** forms a fluid tight seal with an interior surface of chamber **323**, preventing fluid flow through distal opening **390**. In a retracted state (shown in FIG. 3B), biasing member **331** may exert a force on rod **325** pushing rod distally towards distal opening **390**. Proximal end **329** of rod **325** may contact slider **321** and counteract the force exerted on rod **325** from biasing member **331**. In some examples, when a user transitions proximal end **329** from a position contacting first portion **355** of slider **321** to a position contacting second portion **357** of slider **321**, proximal end **329** may slide across second portion **357**, and rod **325** may move distally towards distal opening **390** via a force exerted on rod **325** from biasing member **331**.

[0037] Slider **321** may extend across cover **333** and across a proximal portion of chamber **323**. Slider **321** may be rectangular or any other suitable geometric shape. A proximal facing surface of slider **321** may include a first portion **355** and a second portion **357**. In some examples, first portion **355** may be planar and second portion **357** may be curved. Slider **321** may vary in thickness with a larger thickness at a portion of slider **321** including the first portion **355** and a smaller thickness at a portion of slider **321** including a second portion **357**. For example, slider **321** may extend outward from a central longitudinal axis of slider **321** as slider **321** extends from the second portion **357** to

the first portion 355. A distal facing surface 376 of slider 321 may be planar and may be configured to slide across cover 333 and/or a proximal end of chamber 323. FIGS. 4A and 4B are top views of valve assembly 300 showing chamber 323, cover 333, proximal end 329 of rod 325, and slider 321 including opening 371. FIG. 4A corresponds to the position of valve assembly 300 shown in FIG. 3A, and FIG. 4B corresponds to the position of valve assembly 300 shown in FIG. 3B. Slider 321 may be designed with a slope of second portion 357 such that when slider 321 is translated from a position in which proximal end 329 moves from a position contacting first portion 355 to a position contacting second portion 357, biasing member 331 returns rod 325 to a closed position, which may facilitate dosing of agent 339.

[0038] Slider 321 is configured to translate transverse to the central longitudinal axis of rod 325 and is configured to move rod 325 in a proximal direction or a distal direction (up or down in the Figures). As slider 321 moves in a direction transverse to the longitudinal axis of rod 325, proximal end 329 slideably engages proximal facing surfaces 355, 357 of slider 321. For example, a user may transition valve assembly 300 from a first state shown in FIGS. 3A and 4A to a second state shown in FIGS. 3B and 4B by translating slider across cover 333 such that proximal end 329 slides across second portion 357 to first portion 355, and thus forcing rod 325 to move proximally via the interaction between proximal end 329 and proximally facing surfaces 355, 357 of slider 321. In some examples, slider 321 may include a flange 373 extending radially outward from a longitudinal axis of slider 321. Flange 373 may be configured to prevent movement of slider 321 across cover 333 once flange 373 engages an exterior surface of chamber 323. FIG. 3B shows an example of a position of valve assembly 300 in which flange 373 engages an exterior surface of chamber 323. Slider 321 may be configured to couple to an actuator to facilitate movement of slider 321 to actuate valve assembly 300. For example, slider 321 may be coupled to a trigger assembly (such as a trigger similar to trigger 574 shown in FIGS. 5A, 5B, and 8).

[0039] In operation, a user may operate valve assembly 300 while operating delivery system 200, with valve assembly 300 serving as valve 220 shown in FIG. 2. Once a fluid container 206 is coupled to input 209 and supplies pressurized fluid to delivery system 200, a user may actuate valve assembly 300 to selectively release agent 339 through valve assembly 300. For example, a user may translate slider 321 in a first direction to transition valve assembly from a closed state (shown in FIG. 3A) to an open state (shown in FIG. 3B) to release agent 339 through output channel 353 to be delivered to target tissue of a patient. Agent 339 may be moved from container 204, through catheter 202, and into valve assembly 300, and output through output channel 353 for delivery to distal opening 215. Agent 339 may be moved by a force applied to agent 339 via pressurized fluid supplied by fluid container 206. To stop the release of agent 339 through output channel 353, a user may translate slider 321 in a second direction opposite the first direction to move proximal end 329 from a position contacting first portion 355 of a proximally-facing surface of slider 321 to a position contacting second portion 357 of a proximally-facing surface of slider 321, thus allow biasing member 331 to move rod 325 distally and move plunger 327 into a position fluidically sealing distal opening 390. Once plunger 327 is at a position fluidically sealing distal opening 390, fluid flow (shown as arrow 341, 343 and 345 in FIGS. 3A and 3B) is prevented from reaching output channel 353. In some examples, the user may transition slider 321 from the position shown in FIG. 4B to the position shown in FIG. 4A to stop the release of agent 339 from output channel 353.

[0040] FIGS. 5A and 5B show an alternative embodiment of a valve assembly 500 that utilizes a cam and follower mechanism to actuate the valve assembly 500. Valve assembly 500 may be used as valve 220 in FIG. 2. In some examples, valve assembly 500 may be configured to operate under a constant fluid pressure feed into an input channel 551. Valve assembly 500 may include a valve chamber 523, an input channel 551 including input lumen 550, an output channel 553 including output lumen 552, a rod 525 including a plunger 527 at a distal end of rod 525, a biasing member 531, a cover 533, and a coupler 537. Any of the components of valve assembly 500 may have any

of the attributes and/or characteristics of components of valve assembly **300**. Valve assembly **500** may be contained within a holding apparatus (not shown) that may be any suitable shape. For example, valve assembly **500** may be incorporated into a handle for a user to hold during operation (shown in FIG. **8**).

[0041] Rod **525** may include a first portion **526**, a second portion **528**, a third portion **530**, and a wheel **560** at a distal end of third portion **530**. Each of first portion **526**, second portion **528**, and third portion **530** may be rigid and cylindrical or any other suitable shape. In some examples, wheel **560** may be rotatable relative to third portion **530**. In other examples, wheel **560** may be fixedly coupled to third portion **530**, may be configured to slidably engage cam **571**, and may not rotate. Although third portion **530** and input channel **551** are both shown in cross-section, third portion **530** is in a different plane than input channel **551** such that third portion **530** and input channel **551** do not intersect, but extend transverse to each other. First portion **526** may extend from plunger **527**, through an opening in cover **533**, to a position proximal to cover **533**. First portion **526** may be fixedly coupled to second portion **528**, and, in some examples, second portion may extend transverse to first portion **526**. In some examples, second portion **528** may have a central longitudinal axis perpendicular to a central longitudinal axis of first portion **526** and/or a central longitudinal axis of third portion **530**. Second portion **528** may be fixedly coupled to third portion **530** at a first end, and may be fixedly coupled to first portion **526** at a second end opposing the first end. Third portion **530** may extend distally from a first end to a second end, with the second end of third portion **530** coupled to wheel **560**. In some examples, a central longitudinal axis of third portion **530** may be parallel to a central longitudinal axis of first portion **526**. In some examples, first portion **526**, second portion **528**, and third portion **530** may form a U-shape. Third portion **530** may be held by a bracket **559** coupled to a housing (not shown) containing valve assembly **500**. Bracket **559** may allow proximal and distal movement of third portion **530**, and may prevent lateral movement transverse to the proximal and distal directions. In some examples, bracket **559** may be integrally formed with and/or fixedly coupled to housing **840** (shown in FIG. **8**).

[0042] Cam **571** may be fixedly coupled to a pinion **572**. Cam **571** may be pear shaped (shown in FIGS. **5A** and **5B**), snail shaped, circular with a pinion coupled at an off-center position, or any other suitable shape, and may be made of any suitable material known in the art. Pinion **572** may be cylindrical, and may be rotatably coupled to a housing (not shown) containing valve assembly **500**. Pinion **572** may include gears (teeth) **579**, and rotation of pinion **572** may result in rotation of cam **571**. Gears **579** may be configured to mate with gears (teeth) **573** of rack **570**. Rack **570** may be coupled to an actuator for translating rack **570** to move gears **573**, such as trigger **574** shown in FIGS. **5A** and **5B**. Alternatively, rack **570** could be actuated via a button, lever, electronically controlled motor, or any other means known in the art. In some examples, movement of rack **570** via actuation of trigger **574** may engage gears **573** with gears **579**, and thus may cause rotation of pinion **572** and cam **571**. A central longitudinal axis of rack **570** may extend transverse to an axis of rotation of pinion **572**. In some examples, rotation of pinion **572** via rack **570** may transition cam **571** from a position in which a central longitudinal axis of cam **571** is transverse to a central longitudinal axis of third portion **530** (FIG. **5A**) to a position in which a central longitudinal axis of cam **571** is aligned with a central longitudinal axis of third portion **530** (FIG. **5B**).

[0043] In operation, a user may actuate rack **570** in order to open valve assembly **500** and release agent **539** to a distal opening **215** of catheter **202** (shown in FIG. **2**). After fluid container **206** is coupled to regulator **207**, pressurized fluid may flow into input channel **551** and chamber **523**. In a closed configuration (shown in FIG. **5B**), plunger **527** prevents fluid flow through distal opening **590**. To open valve assembly **500**, the user may actuate rack **570**, for example by pushing on trigger **574**. By pushing on trigger **574**, gears **573** of rack **570** may engage gears **590** of pinion **572**, rotating pinion **572** and cam **571**. As cam **571** rotates, wheel **560** engages the exterior surface of cam **571**, and cam **571** pushes rod **525** proximally. By pushing rod **525** proximally, plunger **527** disengages with the interior surface of chamber **523** and allows fluid to flow through distal opening

590. When the user releases trigger **574**, biasing member **531** may move rod **525** distally and return valve assembly **500** to a closed position (FIG. **5B**).

[0044] FIGS. **6A** and **6B** show an alternative embodiment of a valve assembly **600**. Valve assembly **600** may be used as valve **220** in FIG. **2**. Valve assembly **600** may include a valve chamber **623** including interior portion **635**, an input channel **651** including input lumen **650**, an output channel **653** including output lumen **652**, a cover **633**, and a butterfly valve **601**. Any of the components of valve assembly **600** may have any of the attributes and/or characteristics of components of valve assemblies **300** and **500**. Valve assembly **600** may be contained within a holding apparatus (not shown) that may be any suitable shape. For example, valve assembly **600** may be incorporated into a handle for a user to hold during operation (shown in FIG. **8**). In some examples, valve assembly **600** may be configured to operate under a constant fluid pressure feed into input channel **651**.

[0045] Input channel **651** may extend through an opening in cover **633** and within chamber **623**. For example, a distal portion **624** of input channel **651** may extend along a central longitudinal axis of chamber **623** and a distal end **692** of input channel **651** may, in some examples, be positioned directly above (proximal to) distal opening **690**. Butterfly valve **601** may be oval shaped, or may be any other suitable shape. Butterfly valve **601** may be configured to extend across distal opening **690** so as to prevent fluid flow (shown as arrows **641**, **642** in FIG. **6A**) through distal opening **690** and into lumen **652** of output channel **653**. Butterfly valve **601** may include a pivot rod **602**, and butterfly valve **601** may pivot about pivot rod **602**. In some examples, pivot rod **602** may be configured to be actuated by a user by rotating pivot rod **602** to move butterfly valve **601**. In other examples, valve assembly **600**, and in some embodiments a butterfly valve **601**, may be actuated via a trigger rack and pinion, by gas flow, by a manual twist action, or a by pneumatic liquid/gas system. In some examples, actuating valve assembly **600** to transition valve assembly **600** from a closed to an open configuration may require rotation of butterfly valve **601** ninety degrees. FIG. **6A** shows valve assembly **600** in a closed configuration in which fluid flow, illustrated by arrows **641**, **642**, flows into chamber **623** but is prevented from flowing into lumen **652** of output channel **653**. When valve assembly **600** is in a closed configuration as shown in FIG. **6A**, butterfly valve **601** spans across distal opening **690** and seals distal opening **690**, preventing fluid flow into output channel **653**. In a closed configuration, a longitudinal axis of butterfly valve **601** may extend across distal opening **690**. When valve assembly **600** is in an open configuration as shown in FIG. **6B**, butterfly valve **601** is positioned to allow fluid flow through distal opening **690**, and thus allow agent **639** to flow through output channel **653**, illustrated by arrows **643**, **644**, **695**. For example, in an open configuration, a longitudinal axis of butterfly valve **601** may align with a longitudinal axis of distal portion **624** or a longitudinal axis of chamber **623**. A user may rotate pivot rod **602** to transition valve assembly **600** from a closed configuration to an open configuration, and vice versa. [0046] FIGS. **7A** and **7B** show another alternative embodiment of a valve assembly **700**. Valve assembly **700** may be used as valve **220** in FIG. **2**. Valve assembly **700** may include a valve chamber **723**, an input channel **751** including input lumen **750** and distal end **792**, an output channel **753** including output lumen **752**, a cover **733**, and a ball valve **711**. Any of the components of valve assembly **700** may have any of the attributes and/or characteristics of components of valve assemblies **300**, **500**, and **600**. Valve assembly **700** may be contained within a holding apparatus (not shown) that may be any suitable shape. For example, valve assembly **700** may be incorporated into a handle for a user to hold during operation (shown in FIG. **8**). In some examples, valve assembly **700** may be configured to operate under a constant fluid pressure feed into input channel **751**.

[0047] Ball valve **711** may be configured to extend across distal opening **712** so as to prevent fluid flow (shown as arrow **741**, **742** in FIG. **7A**) through distal opening **712** and into lumen **752** of output channel **753**. Ball valve **711** may be spherical and may include a lumen **713** extending through ball valve **711**. In some examples, ball valve **711** may include a pivot rod (not shown), and ball valve **711** may pivot about pivot rod. In some examples, actuation of a pivot rod may rotate

ball valve **711**. FIG. 7A shows valve assembly **700** in a closed configuration in which fluid flow, illustrated by arrows **741**, **742**, flows into chamber **723** but is prevented from flowing into lumen **752** of output channel **753**. When valve assembly **700** is in a closed configuration as shown in FIG. 7A, ball valve **711** spans across distal opening **712** and seals distal opening **712**, preventing fluid flow into output channel **753**. In a closed configuration, a longitudinal axis of lumen **713** may extend across distal opening **712**, and an interior surface of chamber **723** may cover each end of lumen **713**. When valve assembly **700** is in an open configuration as shown in FIG. 7B, ball valve **711** is positioned to allow fluid flow, illustrated by arrows **743**, **744**, **745**, through lumen **713** and distal opening **712**, and thus allow agent **739** to flow through output channel **753**. For example, in an open configuration, a longitudinal axis of lumen **713** may align with a longitudinal axis of distal portion **724** of input channel **751** or a longitudinal axis of chamber **723**. A user may rotate ball valve **711** to transition valve assembly **700** from a closed configuration to an open configuration, and vice versa. In some examples, actuating valve assembly **700** to transition valve assembly **700** from a closed to an open configuration may require rotation of ball valve **711** ninety degrees. Valve assembly **700**, and in some embodiments a ball valve **711**, may be actuated via a trigger rack and pinion, by gas flow, by a manual twist action, a by pneumatic liquid/gas system, or by any other method known in the art.

[0048] FIG. **8** shows a side cross-sectional view of an exemplary housing **840** containing valve assembly **500**. Housing **840** may include a handle portion **850** configured for a user to hold housing **840**. In some examples, handle portion **850** may be configured to enclose a regulator **851** and/or a fluid container **852**. A user may access an interior portion of handle portion **850** via a cap **853** removably coupled to a portion of handle portion **850**. A trigger **574** may extend outward from housing **840** to allow a user to actuate trigger **574**. Pressurized fluid may flow (shown as arrows **860**, **861**) from fluid container **852**, through regulator **851**, and into valve assembly **500**. Any of the valve assemblies **300**, **500**, **600**, **700** disclosed herein may be incorporated into housing **840**. By providing a handle portion **850**, housing **840** may facilitate the use of valve assembly **300**, **500**, **600**, **700** and provide a more ergonomic way for a user to hold and actuate a valve assembly **300**, **500**, **600**, **700**.

[0049] Unless described otherwise, the structural elements of valve assemblies **300**, **500**, **600**, **700** may be any material known in the art, including but not limited to a metal alloy, a ceramic, and/or a resin.

[0050] It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed device without departing from the scope of the disclosure. Other embodiments of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the invention being indicated by the following claims.

Claims

1. A medical device, comprising: a valve chamber including a first opening and a second opening; an inlet channel fluidly coupled to the valve chamber via the first opening; an outlet channel fluidly coupled to the valve chamber via the second opening; and a rod including a plunger movably disposed within the valve chamber, wherein the rod is configured to selectively seal the valve chamber in response to moving the plunger into the second opening, thereby closing the outlet channel from the valve chamber.
2. The medical device of claim 1, wherein the inlet channel is configured to deliver a fluid into the valve chamber via the first opening, the first opening is positioned relatively above the second opening such that the fluid received in the valve chamber is guided towards the second opening at least partially via gravity.

3. The medical device of claim 1, wherein the plunger is configured to form a fluid-tight seal with the second opening in response to the rod translating distally within the valve chamber and towards the second opening until the plunger contacts an interior surface of the valve chamber.
4. The medical device of claim 3, wherein the interior surface of the valve chamber is tapered towards the second opening, and the plunger includes a shape that is angled to interface with the interior surface.
5. The medical device of claim 1, further comprising: a biasing member coupled to the rod, wherein the biasing member is configured to bias the rod towards the second opening, thereby moving the plunger into the second opening to close the outlet channel from the valve chamber.
6. The medical device of claim 5, wherein the biasing member is a spring that extends circumferentially around the rod, the spring is configured to move between an extended state and a compressed state.
7. The medical device of claim 6, wherein in the extended state, the biasing member is configured to exert a distal force on the rod such that the plunger forms a fluid-tight seal with an interior surface of the valve chamber, thereby preventing fluid flow from the valve chamber into the outlet channel via the second opening.
8. The medical device of claim 6, wherein in the compressed state, the plunger is configured to exert a proximal force on the biasing member such that the plunger removes a fluid-tight seal with an interior surface of the valve chamber, thereby allowing fluid flow from the valve chamber into the outlet channel via the second opening.
9. The medical device of claim 1, further comprising a slidable actuator configured to translate relative to the valve chamber to move the rod and the plunger within the valve chamber.
10. The medical device of claim 9, wherein the slidable actuator is configured to translate outside of the valve chamber in a direction that is transverse to a movement of the rod within the valve chamber.
11. The medical device of claim 9, wherein the slidable actuator includes a slot and the rod extends out of the valve chamber and through the slidable actuator via the slot; and wherein the rod is configured to slide along the slidable actuator to counteract a biasing force exerted on the rod from a biasing member coupled to the rod within the valve chamber.
12. The medical device of claim 9, wherein the slidable actuator is configured to transition the medical device towards a closed state by translating the rod relative to the valve chamber towards a first position in which the plunger engages the second opening.
13. The medical device of claim 12, wherein the slidable actuator is configured to transition the medical device towards an open state by translating the rod relative to the valve chamber towards a second position in which the plunger disengages the second opening.
14. A medical device, comprising: a valve chamber fluidly coupled to each of an inlet via a first opening and an outlet via a second opening; and a rod disposed within the valve chamber between the inlet and the outlet, the rod including a plunger that is configured to move relative to the second opening; wherein the rod is configured to exert a force against the plunger, thereby pushing the plunger towards the second opening of the valve chamber to selectively seal the valve chamber from the outlet when the plunger abuts against the second opening.
15. The medical device of claim 14, wherein the valve chamber is configured to store an agent, the inlet is configured to deliver a fluid into the valve chamber via the first opening, and the outlet is configured to receive a mixture of the agent and the fluid upon the plunger moving away from the second opening to selectively unseal the valve chamber from the outlet when the plunger separates from the second opening.
16. The medical device of claim 14, further comprising: a biasing member coupled around the rod and abutting against the plunger, wherein the biasing member is configured to push the rod towards the second opening, thereby exerting the force against the plunger.
17. The medical device of claim 16, wherein the biasing member is configured to move towards an

extended state in which the biasing member moves the rod downwards relative to the valve chamber to exert the force against the plunger, and the plunger is configured to form a fluid-tight seal against the second opening such that the valve chamber is fluidly decoupled from the outlet.

18. The medical device of claim 17, wherein the biasing member is configured to move towards a compressed state in which the rod moves upwards relative to the valve chamber, and the plunger is configured to remove the fluid-tight seal against the second opening such that the valve chamber is fluidly coupled to the outlet.

19. A medical device, comprising: a valve chamber storing an agent, the valve chamber including a first opening that is configured to receive a fluid in the valve chamber and a second opening that is configured to deliver a mixture of the agent and the fluid out of the valve chamber; an inlet channel in fluid communication with the valve chamber via the first opening, the inlet channel is configured to deliver the fluid towards the first opening; an outlet channel in fluid communication with the valve chamber via the second opening, the outlet channel is configured to deliver the mixture out of the valve chamber via the second opening; and a rod configured to translate within the valve chamber between the inlet channel and the outlet channel to selectively control release of the mixture from the valve chamber towards the outlet channel by disengaging the second opening.

20. The medical device of claim 19, wherein the rod is configured to fluidly decouple the valve chamber from the outlet channel by closing the second opening to form the mixture in the valve chamber; and wherein the rod is configured to fluidly couple the valve chamber with the outlet channel by opening the second opening to deliver the mixture out of the valve chamber.
